

Spectral Sequences in Practice, and the Atiyah–Hirzebruch Torsion Counterexample to the Integral Hodge Conjecture

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1 Conventions on spectral sequences

A (cohomological, first–quadrant) spectral sequence is a sequence of bigraded abelian groups $\{E_r^{p,q}\}$ together with differentials

$$d_r: E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}, \quad d_r \circ d_r = 0,$$

such that $E_{r+1}^{p,q} = \ker d_r / \operatorname{im} d_r$ at (p, q) . We say it *converges* to a graded group $H^\bullet$, written $E_2^{p,q} \Rightarrow H^{p+q}$, if H^n carries a finite decreasing filtration $H^n = F^0 \supseteq F^1 \supseteq \dots \supseteq F^n \supseteq F^{n+1} = 0$ with

$$E_\infty^{p,q} \cong F^p H^{p+q} / F^{p+1} H^{p+q}.$$

Three separate pieces of information are always at play, and it is worth naming them because *each* is a distinct step in a computation:

- (D) **Differentials.** One must determine the maps d_r . Often they are forced by knowing the abutment (as in Part I), or identified with a known cohomology operation (as in Part II).
- (M) **Multiplicativity.** In good cases the E_r are bigraded rings, d_r is a derivation $d_r(xy) = d_r(x)y + (-1)^{|x|}x d_r(y)$ (with $|x| = p + q$), and the filtration on $H^\bullet$ is multiplicative. This propagates a single differential across the whole page.
- (E) **The extension problem.** Even after computing E_∞ , one only knows the *associated graded* $\operatorname{gr} H^n = \bigoplus_p E_\infty^{p,n-p}$. Recovering H^n requires solving extensions $0 \rightarrow F^{p+1} \rightarrow F^p \rightarrow E_\infty^{p,n-p} \rightarrow 0$, which can be nonsplit.

Parts I and III (the $\mathbb{RP}^{2n}$ example) are chosen precisely so that you meet all three phenomena.

2 A fully worked example: the Serre SS of a path–loop fibration

2.1 The setup

Fix $n \geq 2$ and let $PS^n = \{\gamma: [0, 1] \rightarrow S^n : \gamma(0) = x_0\}$ be the space of based paths, with the fibration

$$\Omega S^n \hookrightarrow PS^n \xrightarrow{\operatorname{ev}_1} S^n, \quad \operatorname{ev}_1(\gamma) = \gamma(1),$$

whose fibre is the based loop space ΩS^n . The total space PS^n is *contractible* (contract each path to the constant path), so

$$H^m(PS^n; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & m = 0, \\ 0, & m > 0. \end{cases} \quad (1)$$

Since S^n is simply connected for $n \geq 2$, the local system on the base is trivial and the cohomology Serre spectral sequence reads

$$E_2^{p,q} = H^p(S^n; H^q(\Omega S^n; \mathbb{Z})) \implies H^{p+q}(PS^n; \mathbb{Z}).$$

Write $A^q := H^q(\Omega S^n; \mathbb{Z})$; this is what we want to compute. Because $H^p(S^n; \mathbb{Z}) = \mathbb{Z}$ for $p \in \{0, n\}$ and 0 otherwise, *only two columns are nonzero*:

$$E_2^{0,q} = A^q, \quad E_2^{n,q} = A^q, \quad E_2^{p,q} = 0 \text{ for } p \neq 0, n.$$

2.2 Which differentials can be nonzero?

A differential $d_r: E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}$ can be nonzero only if both columns p and $p+r$ are nonzero, i.e. only if $\{p, p+r\} = \{0, n\}$. Hence *the only possibly nonzero differential is d_n* , and

$$E_2 = E_3 = \dots = E_n, \quad E_{n+1} = E_\infty.$$

The single relevant family of maps is the *transgression*

$$d_n: E_n^{0,q} = A^q \rightarrow E_n^{n, q-n+1} = A^{q-n+1}.$$

2.3 Convergence forces d_n to be an isomorphism

By (1) the abutment is trivial except in total degree 0, so $E_\infty^{p,q} = 0$ unless $(p,q) = (0,0)$. Since $E_{n+1} = E_\infty$, the only group that may survive is $E_\infty^{0,0} = \mathbb{Z}$.

Low degrees. For $0 < q < n-1$ the group $E_n^{0,q} = A^q$ receives no differential (its potential source $E_n^{n,q-n+1}$ sits in negative fibre degree) and maps nowhere nontrivially; hence it survives to E_∞ and must vanish:

$$A^q = 0 \quad (1 \leq q \leq n-2).$$

Transgression. In degree $q = n-1$ the class $s \in H^n(S^n) = E_n^{n,0}$ must die. The only differential that can kill it is

$$d_n: E_n^{0,n-1} = A^{n-1} \xrightarrow{\cong} E_n^{n,0} = \mathbb{Z}\langle s \rangle.$$

For $E_\infty^{n,0} = 0$ this map must be surjective; for $E_\infty^{0,n-1} = 0$ it must be injective. Therefore $A^{n-1} = \mathbb{Z}$, generated by a class a with

$$d_n(a) = s. \quad (2)$$

Bootstrapping. More generally, for every q the map $d_n: A^q \rightarrow A^{q-n+1}$ must be an isomorphism whenever both are off the corner (source in column 0, target in column n , and neither is $E_\infty^{0,0}$). Feeding in $A^0 = \mathbb{Z}$, $A^q = 0$ for $1 \leq q \leq n-2$, and iterating $A^q \cong A^{q-(n-1)}$ gives:

$$H^q(\Omega S^n; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & q \equiv 0 \pmod{n-1}, \quad q \geq 0, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Denote the generator of $H^{k(n-1)}$ by a_k , with $a_0 = 1$ and $a_1 = a$.

2.4 A concrete page: $n = 3$

For S^3 the fibre cohomology lives in even degrees $0, 2, 4, 6, \dots$, and the $E_2 = E_3$ page has just the two columns $p = 0, 3$:

$q = 6$	$\mathbb{Z}\langle a_3 \rangle$	0	0	$\mathbb{Z}\langle s a_3 \rangle$
$q = 4$	$\mathbb{Z}\langle a_2 \rangle$	0	0	$\mathbb{Z}\langle s a_2 \rangle$
$q = 2$	$\mathbb{Z}\langle a_1 \rangle$	0	0	$\mathbb{Z}\langle s a_1 \rangle$
$q = 0$	$\mathbb{Z}\langle 1 \rangle$	0	0	$\mathbb{Z}\langle s \rangle$
	$p = 0$	1	2	$p = 3$

The differential d_3 points three columns to the right and two rows down, $d_3: E_3^{0,2k} \rightarrow E_3^{3,2k-2}$, i.e.

$$a_k \longmapsto (\text{integer}) \cdot s a_{k-1}.$$

Convergence (everything off the corner must die) forces each such d_3 to be an *isomorphism* $\mathbb{Z} \xrightarrow{\cong} \mathbb{Z}$; after fixing signs we may take

$$d_3(a_k) = s a_{k-1} \quad (k \geq 1). \quad (4)$$

2.5 Multiplicativity and the integral extension problem

So far (3) is purely additive. Now use that the Serre spectral sequence is *multiplicative*: $E_2^{\bullet, \bullet} = H^*(S^n) \otimes H^*(\Omega S^n)$ as bigraded rings, and d_n is a derivation. Take $n = 3$, so $a = a_1$ has even total degree and there are no Koszul signs. From (4) and the Leibniz rule,

$$d_3(a_1^2) = d_3(a_1) a_1 + a_1 d_3(a_1) = s a_1 + a_1 s = 2 s a_1.$$

But $a_1^2 \in H^4(\Omega S^3) = \mathbb{Z}\langle a_2 \rangle$, say $a_1^2 = \lambda a_2$, and by (4) $d_3(a_2) = s a_1$. Hence

$$2 s a_1 = d_3(a_1^2) = \lambda d_3(a_2) = \lambda s a_1 \implies \boxed{a_1^2 = 2 a_2}.$$

This is the crux: the ring is *not* polynomial. Iterating (induction with $a_1 a_{k-1} = k a_k$, obtained the same way) gives

$$a_1^k = k! a_k.$$

Theorem 2.1. *For n odd, $H^*(\Omega S^n; \mathbb{Z})$ is the divided power algebra $\Gamma_{\mathbb{Z}}[a]$ on one generator a of degree $n - 1$: it is free of rank one in each degree $k(n - 1)$ on a class γ_k , with $\gamma_1 = a$, $a^k = k! \gamma_k$, and $\gamma_i \gamma_j = \binom{i+j}{i} \gamma_{i+j}$. In particular it is not a polynomial ring over $\mathbb{Z}$. $\square$*

Remark 2.2. Every technique you need downstream appeared here: **(D)** differentials forced by the abutment via (1); **transgression** (2); **(M)** a single differential propagated by the Leibniz rule; **(E)** an extension/divided-power subtlety visible only integrally ($a_1^2 = 2a_2$). Reducing mod a prime p would hide this: $H^*(\Omega S^3; \mathbb{F}_p)$ is polynomial. The moral — “torsion and integral structure hide in the multiplicative/extension data” — is exactly what powers Part IV.

3 The Atiyah–Hirzebruch spectral sequence

3.1 Definition

Let $E^\bullet$ be a generalized (multiplicative) cohomology theory and X a finite CW complex, filtered by skeleta $X^{(p)}$. The exact couple of the filtration produces the *Atiyah–Hirzebruch spectral sequence* (AHSS)

$$E_2^{p,q} = H^p(X; E^q(\text{pt})) \implies E^{p+q}(X), \quad (5)$$

converging to the filtration $F^p E^n(X) = \ker(E^n(X) \rightarrow E^n(X^{(p-1)}))$. Here $E^q(\text{pt})$ are the coefficient groups. When $E = H(-; A)$ is ordinary cohomology this collapses at E_2 ; the content is entirely in *generalized* theories.

3.2 Complex K –theory

For $E = \text{KU}$ (complex topological K –theory) Bott periodicity gives

$$\text{KU}^q(\text{pt}) = \begin{cases} \mathbb{Z}, & q \text{ even,} \\ 0, & q \text{ odd,} \end{cases}$$

with the Bott class $\beta_{\text{Bott}} \in \text{KU}^{-2}(\text{pt})$ a unit. Hence

$$E_2^{p,q} = \begin{cases} H^p(X; \mathbb{Z}), & q \text{ even,} \\ 0, & q \text{ odd,} \end{cases} \implies \text{KU}^{p+q}(X). \quad (6)$$

The nonzero groups occupy the even rows; the whole picture is 2–periodic in q . Two immediate consequences:

- all *even* differentials vanish, $d_{2k} = 0$, because they map between an even and an odd row (one of which is zero);
- rationally the sequence degenerates, $E_2 \otimes \mathbb{Q} \cong E_\infty \otimes \mathbb{Q}$, so $\mathrm{KU}^0(X) \otimes \mathbb{Q} \cong \bigoplus_k H^{2k}(X; \mathbb{Q})$ (the Chern character). All higher differentials are therefore *torsion* operations.

4 The first differential: $d_3 = \widetilde{\mathrm{Sq}}^3$

Theorem 4.1 (Atiyah–Hirzebruch). *In the AHSS for KU the first potentially nonzero differential is $d_3: E_3^{p,q} \rightarrow E_3^{p+3,q-2}$, and under (6) it is the integral Steenrod operation*

$$d_3 = \widetilde{\mathrm{Sq}}^3 = \beta \circ \mathrm{Sq}^2 \circ \rho : H^p(X; \mathbb{Z}) \rightarrow H^{p+3}(X; \mathbb{Z}),$$

where $\rho: H^*(-; \mathbb{Z}) \rightarrow H^*(-; \mathbb{F}_2)$ is reduction mod 2, Sq^2 is the Steenrod square, and $\beta: H^*(-; \mathbb{F}_2) \rightarrow H^{*+1}(-; \mathbb{Z})$ is the integral Bockstein of $0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{F}_2 \rightarrow 0$. In particular $\widetilde{\mathrm{Sq}}^3$ takes values in 2-torsion.

Remark 4.2. Because $\rho\beta = \mathrm{Sq}^1$, the mod 2 reduction of $\widetilde{\mathrm{Sq}}^3$ is the genuine Steenrod square $\mathrm{Sq}^3 = \mathrm{Sq}^1 \mathrm{Sq}^2$. This is the fact we compute with in Part IV. Higher differentials $d_5, d_7, \dots$ exist and are secondary/tertiary operations, but the counterexample only needs d_3 .

5 A second AHSS worked example: $\widetilde{\mathrm{KU}}^0(\mathbb{RP}^{2n}) = \mathbb{Z}/2^n$

This example isolates the *extension problem* in the AHSS (here every differential vanishes, yet the answer is a nonsplit extension).

The integral cohomology of $X = \mathbb{RP}^{2n}$ is

$$H^p(\mathbb{RP}^{2n}; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & p = 0, \\ \mathbb{Z}/2, & p = 2, 4, \dots, 2n, \\ 0, & p \text{ odd}. \end{cases}$$

Plugging into (6), the E_2 -page (row $q = 0$; all other rows are a 2-periodic copy) has nonzero entries only in *even* columns:

$q = 0$	$\mathbb{Z}$	0	$\mathbb{Z}/2$	0	$\mathbb{Z}/2$	$\cdots$	$\mathbb{Z}/2$
$p = 0$	1	2	3	4	$\cdots$	$2n$	

Now run the checklist:

- **(D) Differentials.** $d_3: H^p \rightarrow H^{p+3}$ maps an even column to an odd column, and $H^{\mathrm{odd}}(\mathbb{RP}^{2n}) = 0$. Hence $d_3 = 0$, and for the same parity reason *all* d_r vanish. The sequence collapses: $E_2 = E_\infty$. (Note $\widetilde{\mathrm{Sq}}^3$ is identically 0 here — this space is *not* a counterexample; contrast Part IV.)
- **(E) Extensions.** The associated graded of $\widetilde{\mathrm{KU}}^0(\mathbb{RP}^{2n})$ is $\bigoplus_{i=1}^n E_\infty^{2i, -2i} = (\mathbb{Z}/2)^\oplus n$, of order 2^n . But the filtration is nonsplit: writing $\eta = \lambda_{\mathbb{C}} - 1$ for the reduced class of the complexified tautological line bundle, one has $\eta^{n+1} = 0$ while η has additive order 2^n , so

$$\widetilde{\mathrm{KU}}^0(\mathbb{RP}^{2n}) \cong \mathbb{Z}/2^n.$$

The successive quotients $F^{2i}/F^{2i+2} = \mathbb{Z}/2$ assemble by *nontrivial* extensions $\mathbb{Z}/2^{i-1} \twoheadrightarrow \mathbb{Z}/2^i \twoheadrightarrow \mathbb{Z}/2$.

This is the same phenomenon as $a_1^2 = 2a_2$ in Part I: E_∞ knows the graded pieces, not the group.

6 The obstruction: algebraic classes are permanent cycles

6.1 Cycle classes lift to K -theory

Let X be a smooth complex projective variety and $Z \subset X$ a closed analytic subset of complex codimension k . The coherent sheaf $\mathcal{O}_Z$ defines a class in K^0 supported on Z ; concretely a locally free resolution of $\mathcal{O}_Z$ gives a class in $KU^0(X)$ lying in the topological filtration step F^{2k} . Under the Chern character / Grothendieck–Riemann–Roch this recovers the cycle class up to a universal factor:

$$c_k(\mathcal{O}_Z) = (-1)^{k-1}(k-1)! [Z] \in H^{2k}(X; \mathbb{Z}). \quad (7)$$

The point is representability: $[Z]$ is (a rational multiple of) the image in $\mathrm{gr}^{2k} KU^0(X) = E_\infty^{2k, -2k}$ of an honest element of $KU^0(X)$.

Theorem 6.1 (Atiyah–Hirzebruch criterion). *For $Z \subset X$ closed analytic of codimension k , the cycle class $[Z] \in H^{2k}(X; \mathbb{Z})$ is a permanent cycle in the AHSS for KU : it lies in the kernel of every differential d_r , $r \geq 2$. In particular*

$$\widetilde{\mathrm{Sq}}^3 [Z] = d_3 [Z] = 0.$$

More generally, all odd-degree Steenrod operations vanish on the classes of algebraic cycles (in $H^(X; \mathbb{Z})$ and in $H^*(X; \mathbb{Z}/p)$).*

Idea. An element of $F^{2k} KU^0(X)$ surviving to $E_\infty^{2k, -2k}$ is by definition in $\ker d_r$ for all r . Since $[Z]$ (up to the invertible-on-the-relevant-page factor $(k-1)!$, which does not affect the vanishing statement 2-locally in the range needed) is the image of $[\mathcal{O}_Z] \in F^{2k} KU^0(X)$, it is a permanent cycle. The extension to all odd Steenrod operations and to $\mathbb{Z}/p$ coefficients is Totaro’s refinement via complex cobordism MU . $\square$

Corollary 6.2 (The contrapositive we exploit). *If $\alpha \in H^{2k}(X; \mathbb{Z})$ satisfies $\widetilde{\mathrm{Sq}}^3 \alpha \neq 0$, then α is not in the image of the cycle class map $\mathrm{CH}^k(X) \rightarrow H^{2k}(X; \mathbb{Z})$; i.e. α is not algebraic.* $\square$

7 The counterexample

7.1 Statement

Definition 7.1. For X smooth projective/ $\mathbb{C}$, the group of *integral Hodge classes* in degree $2k$ is

$$\mathrm{Hdg}^{2k}(X, \mathbb{Z}) = \{ \alpha \in H^{2k}(X; \mathbb{Z}) : \alpha_{\mathbb{C}} \in H^{k,k}(X) \}, \quad \alpha_{\mathbb{C}} = \text{image in } H^{2k}(X; \mathbb{C}).$$

The *Integral Hodge Conjecture* (IHC) asserts that the cycle class map $\mathrm{CH}^k(X) \rightarrow \mathrm{Hdg}^{2k}(X, \mathbb{Z})$ is surjective.

Theorem 7.2 (Atiyah–Hirzebruch, 1962). *The IHC is false for $k \geq 2$. Explicitly, there is a smooth complex projective variety X of dimension 7 carrying a 2-torsion class $\alpha \in H^4(X; \mathbb{Z})$ which is an integral Hodge class but is not the class of any algebraic cycle.*

The proof has three independent ingredients: (i) torsion classes are automatically Hodge; (ii) a construction of a projective X realizing a chosen piece of group cohomology; (iii) an explicit torsion class α with $\widetilde{\mathrm{Sq}}^3 \alpha \neq 0$, defeated by Corollary 6.2.

7.2 Torsion classes are Hodge

Lemma 7.3. *Any torsion class $\alpha \in H^{2k}(X; \mathbb{Z})_{\text{tors}}$ lies in $\text{Hdg}^{2k}(X, \mathbb{Z})$.*

Proof. The map $H^{2k}(X; \mathbb{Z}) \rightarrow H^{2k}(X; \mathbb{C})$ kills torsion, since $H^{2k}(X; \mathbb{C})$ is a $\mathbb{C}$ -vector space. Thus $\alpha_{\mathbb{C}} = 0$, which lies in $H^{k,k}(X)$ trivially. $\square$

So to break the IHC it suffices to produce a *non-algebraic torsion class* of even degree.

7.3 Realizing group cohomology projectively (Godeaux–Serre)

Theorem 7.4 (Serre’s construction; Atiyah–Hirzebruch Prop. 6.6). *Let G be a finite group and $N \in \mathbb{N}$. Choose a faithful complex representation $G \hookrightarrow GL(V)$; then G acts freely on the complement of a closed subvariety of high codimension in a large sum $V^{\oplus m}$, and intersecting with generic hypersurfaces (Lefschetz) yields a smooth projective variety X with a free G -action such that the classifying map $X/G \rightarrow BG$ induces isomorphisms*

$$H^i(BG; \mathbb{Z}) \xrightarrow{\cong} H^i(X/G; \mathbb{Z}) \quad (0 \leq i \leq N).$$

In fact $X/G \rightarrow BG \times BS^1$ is r -connected with $r = \dim_{\mathbb{C}}(X/G)$.

Thus any torsion class in $H^{\leq N}(BG; \mathbb{Z})$ transports to a smooth projective variety, together with all Steenrod operations, provided $\dim_{\mathbb{C}} X \geq N$. For our class $\alpha \in H^4$ with $\widetilde{\text{Sq}}^3 \alpha \in H^7$, we need $N \geq 7$, hence $\dim_{\mathbb{C}} X = 7$ — this is where the “dimension 7” comes from.

7.4 The explicit torsion class on $B(\mathbb{Z}/2)^3$

Take $G = (\mathbb{Z}/2)^3$. With $\mathbb{F}_2$ -coefficients,

$$H^*((\mathbb{Z}/2)^3; \mathbb{F}_2) = \mathbb{F}_2[x_1, x_2, x_3], \quad |x_i| = 1.$$

Let $w = x_1 x_2 x_3 \in H^3$, and let $\alpha = \beta(w) \in H^4((\mathbb{Z}/2)^3; \mathbb{Z})$ be its integral Bockstein; α is 2-torsion, and $\rho(\alpha) = \text{Sq}^1 w$. We show $\widetilde{\text{Sq}}^3 \alpha \neq 0$ by computing its mod 2 reduction.

Reduction to Sq ’s. Using $\rho\beta = \text{Sq}^1$ and Theorem 4.1,

$$\rho(\widetilde{\text{Sq}}^3 \alpha) = \rho\beta \text{Sq}^2 \rho\beta(w) = \text{Sq}^1 \text{Sq}^2 \text{Sq}^1(w).$$

Adem relations. From $\text{Sq}^1 \text{Sq}^2 = \text{Sq}^3$ and $\text{Sq}^3 \text{Sq}^1 = \text{Sq}^2 \text{Sq}^2$ (equivalently $\text{Sq}^2 \text{Sq}^2 = \text{Sq}^3 \text{Sq}^1$),

$$\text{Sq}^1 \text{Sq}^2 \text{Sq}^1 = \text{Sq}^3 \text{Sq}^1 = \text{Sq}^2 \text{Sq}^2.$$

So it suffices to show $\text{Sq}^2 \text{Sq}^2(w) \neq 0$.

The computation. With the total square $\text{Sq}(x_i) = x_i + x_i^2$ and the Cartan formula, the degree-5 part of $\text{Sq}(w)$ is

$$\text{Sq}^2(x_1 x_2 x_3) = x_1^2 x_2^2 x_3 + x_1^2 x_2 x_3^2 + x_1 x_2^2 x_3^2.$$

Applying Sq^2 once more (using $\text{Sq}(x_i^2) = x_i^2 + x_i^4$) to each monomial and adding mod 2:

$$\text{Sq}^2(x_1^2 x_2^2 x_3) = x_1^4 x_2^2 x_3 + x_1^2 x_2^4 x_3 + x_1^2 x_2^2 x_3^2,$$

$$\text{Sq}^2(x_1^2 x_2 x_3^2) = x_1^4 x_2 x_3^2 + x_1^2 x_2^2 x_3^2 + x_1^2 x_2 x_3^4,$$

$$\text{Sq}^2(x_1 x_2^2 x_3^2) = x_1^2 x_2^4 x_3^2 + x_1 x_2^2 x_3^4 + x_1 x_2^2 x_3^2.$$

The three copies of $x_1^2 x_2^2 x_3^2$ add to a single copy (mod 2), so

$$\boxed{\mathrm{Sq}^2 \mathrm{Sq}^2(x_1 x_2 x_3) = x_1^2 x_2^2 x_3^2 + \sum_{6 \text{ perms of } (4,2,1)} x_1^a x_2^b x_3^c}$$

$$= x_1^2 x_2^2 x_3^2 + x_1^4 x_2^2 x_3 + x_1^4 x_2 x_3^2 + x_1^2 x_2^4 x_3 + x_1 x_2^4 x_3^2 + x_1^2 x_2 x_3^4 + x_1 x_2^2 x_3^4.$$

These monomials are distinct and nonzero in the polynomial ring, so $\mathrm{Sq}^2 \mathrm{Sq}^2(w) \neq 0$. Consequently

$$\rho(\widetilde{\mathrm{Sq}}^3 \alpha) = \mathrm{Sq}^1 \mathrm{Sq}^2 \mathrm{Sq}^1(w) = \mathrm{Sq}^2 \mathrm{Sq}^2(w) \neq 0, \quad \text{hence} \quad \widetilde{\mathrm{Sq}}^3 \alpha \neq 0.$$

7.5 Conclusion

Proof of Theorem 7.2. By Theorem 7.4 with $G = (\mathbb{Z}/2)^3$ and $N = 7$, take a smooth projective 7-fold X with $H^i(X; \mathbb{Z}) \cong H^i(B(\mathbb{Z}/2)^3; \mathbb{Z})$ compatibly with Steenrod operations for $i \leq 7$. Let $\alpha \in H^4(X; \mathbb{Z})$ be the image of $\beta(x_1 x_2 x_3)$. Then:

- α is 2-torsion, hence $\alpha \in \mathrm{Hdg}^4(X, \mathbb{Z})$ by Lemma 7.3;
- $\widetilde{\mathrm{Sq}}^3 \alpha \neq 0$ by the computation above (transported from $B(\mathbb{Z}/2)^3$);
- therefore α is not algebraic, by Corollary 6.2.

So α is an integral Hodge class that is not a $\mathbb{Z}$ -combination of algebraic cycle classes: the IHC fails in codimension 2. $\square$

Remark 7.5 (What “core spectral sequence” really means here). The spectral sequence at the heart of the argument is the AHSS (6) for KU. It enters not through a long page-by-page computation but through one structural fact — Theorem 6.1: *algebraic classes are permanent cycles*. The obstruction is then localized in the single differential $d_3 = \widetilde{\mathrm{Sq}}^3$. In this sense the counterexample is a one-differential phenomenon, in contrast to the multi-page bookkeeping of Part I. Totaro (1997) later refined the whole story by replacing KU with complex cobordism MU, factoring the cycle map through $(MU^*(X) \otimes_{MU^*} \mathbb{Z})^{2k}$ and producing sharper (higher-differential) obstructions.

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